Security Assurance for Modern Big Data APIs

Enea Manzi[1], Marco Anisetti[1,2]

enea.manzi@unimi.it
https://eneamanzi.github.io/

[1]Department of Computer Science, Università degli Studi di Milano

[2]Moon Cloud srl

ITADATA 2026

CHANGE: Background

CHANGE: il concetto, lo schema o il contesto di cui parla il lavoro, con i suoi elementi. Meglio una figura che una definizione lunga.

CHANGE: Motivations

CHANGE: il problema, e perché lo stato dell'arte non basta. Meglio mostrarlo su un esempio concreto (una figura di uno scenario).

CHANGE: Our Approach: Overview

CHANGE: una frase sull'approccio e sotto un punto per componente. Le slide dopo corrispondono ai punti, nello stesso ordine: è un indice implicito dei contenuti, non una scaletta.

CHANGE: Our Approach: Details

CHANGE: una slide per componente, dall'alto verso il basso: dove sta nell'approccio complessivo, poi cos'è. Meglio un'immagine, anche animata, di una definizione lunga. Si copia il frame per ogni componente (Details (1), (2), ...).

CHANGE: Example

CHANGE: l'esempio si può raccontare dentro ogni componente, mentre lo presenti, oppure passo passo dopo l'approccio. Dipende dal tempo.

CHANGE: Experiments

CHANGE: obiettivo degli esperimenti e come sono impostati. Se sono tanti, mostra solo il sottoinsieme che fa passare l'intuizione. Una o due slide.

CHANGE: Results

CHANGE: i risultati, per esempio come grafici, con una sola conclusione per grafico.

CHANGE: Conclusions

CHANGE: una riga su cosa è stato fatto, due punti sui vantaggi, poi i lavori futuri.

Thanks! Questions?

CHANGE: author 1

CHANGE: author 1 email

CHANGE: author 1 web page

CHANGE: author 1 LinkedIn

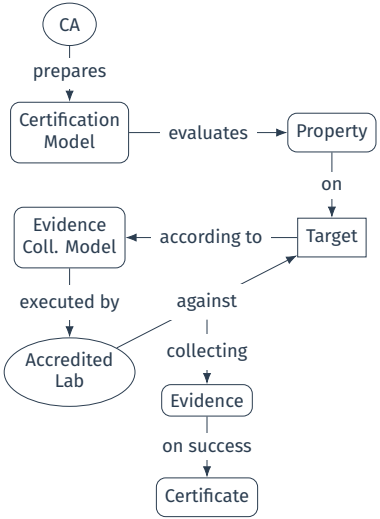
Examples

Ready-to-copy code for the techniques seen in Nicola’s presentations

Example: highlighted list next to a figure

CHANGE: one sentence introducing the concept and its key words, followed by the elements

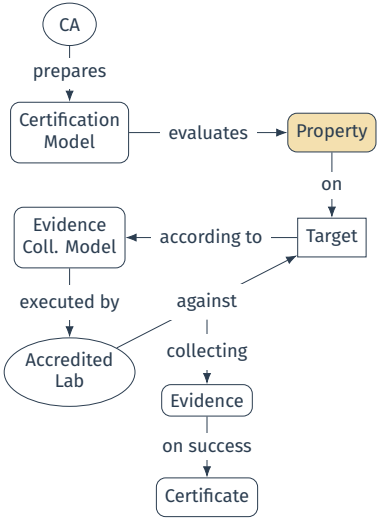
- CHANGE: step 1
- CHANGE: step 2
- CHANGE: step 3
- CHANGE: step 4
- CHANGE: step 5
- CHANGE: step 6



Example: highlighted list next to a figure

CHANGE: one sentence introducing the concept and its key words, followed by the elements

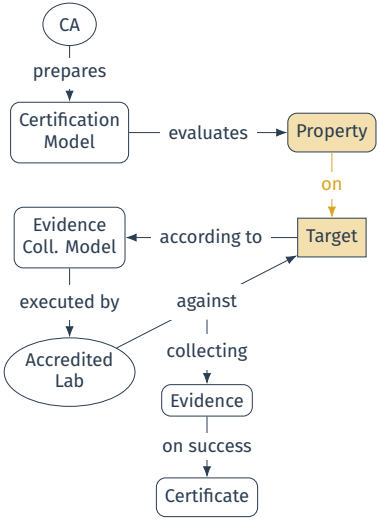
- CHANGE: step 1
- CHANGE: step 2
- CHANGE: step 3
- CHANGE: step 4
- CHANGE: step 5
- CHANGE: step 6



Example: highlighted list next to a figure

CHANGE: one sentence introducing the concept and its key words, followed by the elements

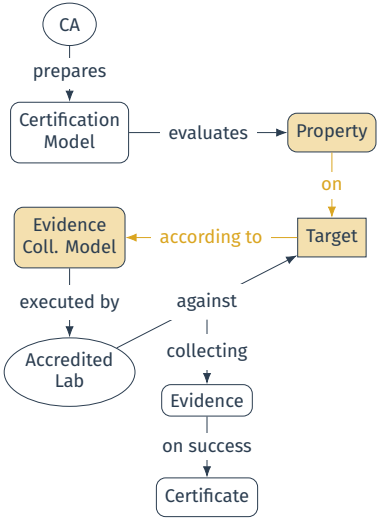
- CHANGE: step 1
- CHANGE: step 2
- CHANGE: step 3
- CHANGE: step 4
- CHANGE: step 5
- CHANGE: step 6



Example: highlighted list next to a figure

CHANGE: one sentence introducing the concept and its key words, followed by the elements

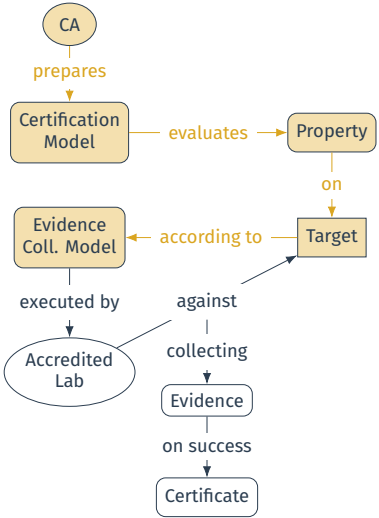
- CHANGE: step 1
- CHANGE: step 2
- CHANGE: step 3
- CHANGE: step 4
- CHANGE: step 5
- CHANGE: step 6



Example: highlighted list next to a figure

CHANGE: one sentence introducing the concept and its key words, followed by the elements

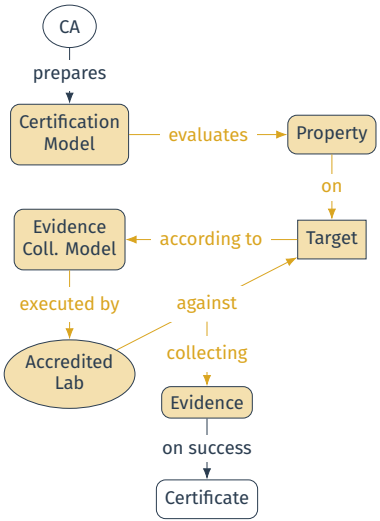
- CHANGE: step 1
- CHANGE: step 2
- CHANGE: step 3
- CHANGE: step 4
- CHANGE: step 5
- CHANGE: step 6



Example: highlighted list next to a figure

CHANGE: one sentence introducing the concept and its key words, followed by the elements

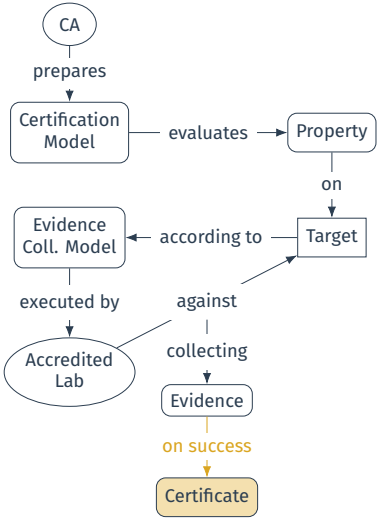
- CHANGE: step 1
- CHANGE: step 2
- CHANGE: step 3
- CHANGE: step 4
- CHANGE: step 5
- CHANGE: step 6



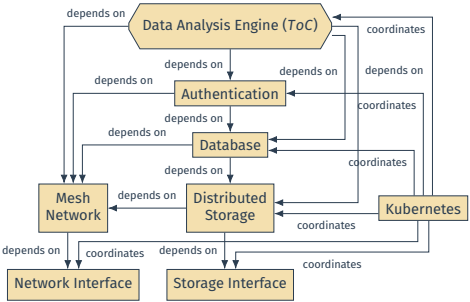
Example: highlighted list next to a figure

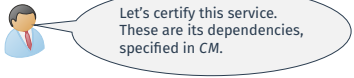
CHANGE: one sentence introducing the concept and its key words, followed by the elements

- CHANGE: step 1
- CHANGE: step 2
- CHANGE: step 3
- CHANGE: step 4
- CHANGE: step 5
- CHANGE: step 6

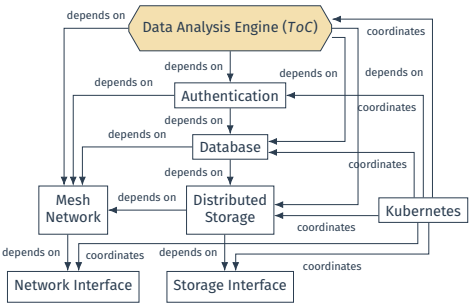


Example: animated scenario figure



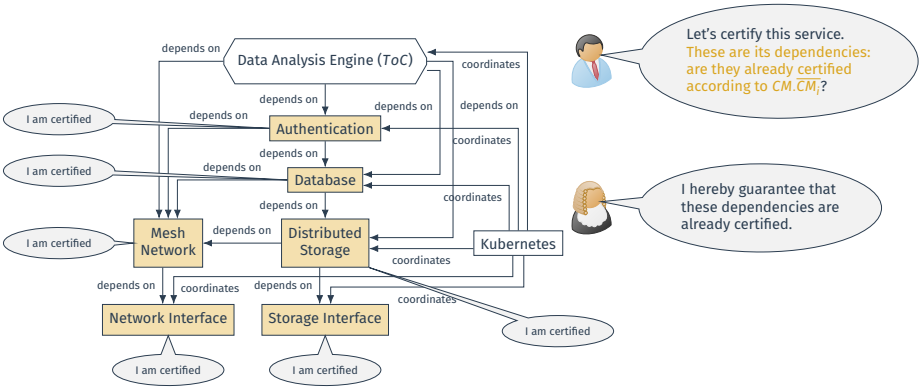


Example: animated scenario figure

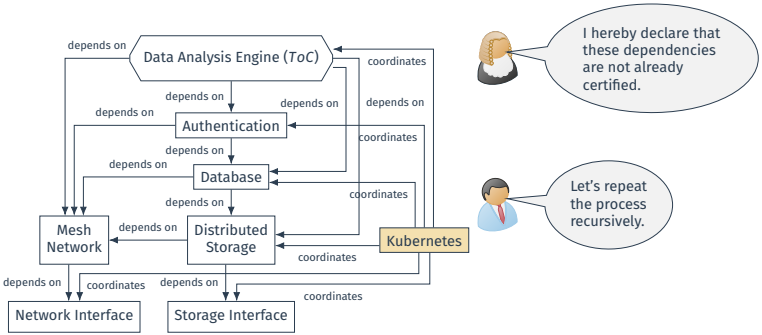




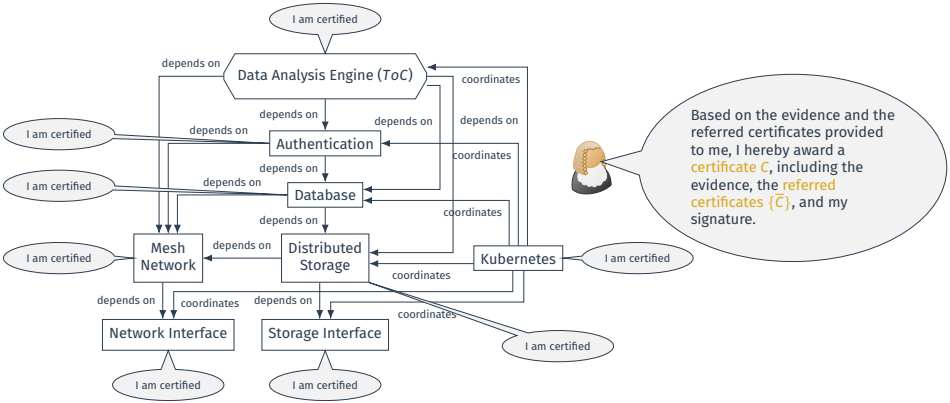
Example: animated scenario figure



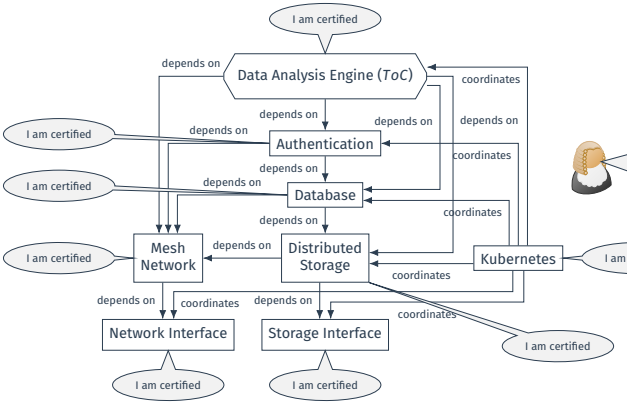
Example: animated scenario figure



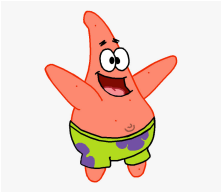
Example: animated scenario figure



Example: animated scenario figure



Based on the evidence and the referred certificates provided to me, I hereby award a certificate C, including the evidence, the referred certificates $\{\tilde{C}\}$, and my signature.



Example: approach overview

Main idea: CHANGE: one sentence stating the idea of the approach

- CHANGE: aspect 1 of the idea, with the key words highlighted
- CHANGE: aspect 2 of the idea
- CHANGE: aspect 3 of the idea

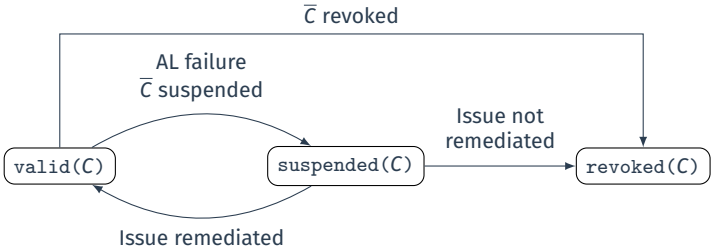
Example: approach details

CHANGE: element name

A CHANGE: element is a tuple $\langle x_1, \{y_1, \ldots, y_n\}, \overline{X}, \texttt{sig} \rangle$, where

- x_1 is CHANGE: description of element 1
- $\{y_i\}$ is a set of *CHANGE: elements*, CHANGE: description of element 2
- $\overline{X}=\{\overline{x}_1, ..., \overline{x}_n\}$ is a set of *referred elements*, each $\overline{x}_i$ CHANGE: relation to the element (CHANGE: description of element 3)
- `sig` is CHANGE: description of element 4.

Example: state diagram



Example: experiments, text and image

Goal: CHANGE: goal of the experiments

- CHANGE: item 1: CHANGE: description of the first point...

Example: experiments, text and image

Goal: CHANGE: goal of the experiments

- CHANGE: item 1: CHANGE: description of the first point...
- ...CHANGE: item 2, which appears at the next step



Example: experiments, changing content

CHANGE: fixed header: CHANGE: sentence that stays at every step, with key words and the symbol T

CHANGE: introduction of the list, in terms of

- CHANGE: point 1
- CHANGE: point 2
  - CHANGE: sub-point of point 2, with an *emphasized* word



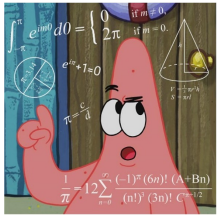
CHANGE: closing sentence

Example: experiments, changing content

CHANGE: fixed header: CHANGE: sentence that stays at every step, with key words and the symbol T

CHANGE: sentence about the key concept T

- CHANGE: point 1
- CHANGE: point 2

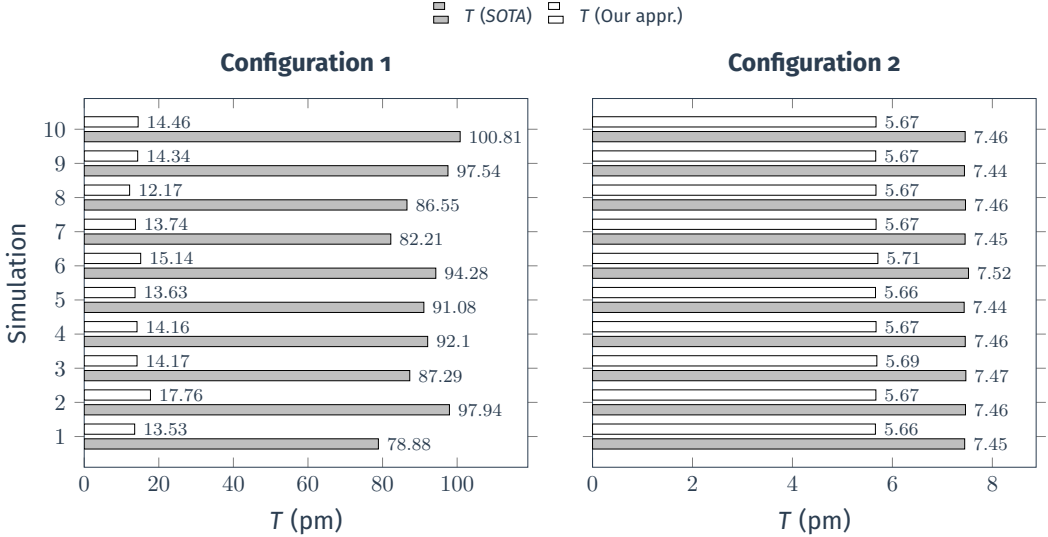


Example: experiments, changing content

CHANGE: fixed header: CHANGE: sentence that stays at every step, with key words and the symbol T

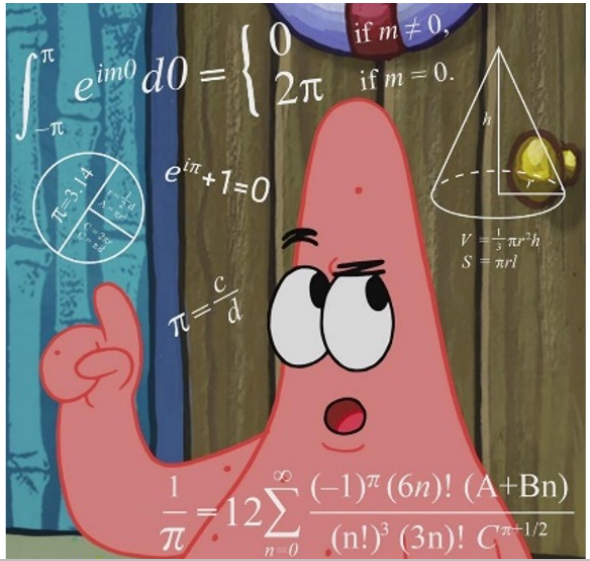
CHANGE: row label	CHANGE: group A		CHANGE: group B		CHANGE: group C	
	CHANGE: A1	CHANGE: A2	CHANGE: B1	CHANGE: B2	CHANGE: C1	CHANGE: C2
1	0	0	0	0	4	16
2	80	20	0	0	0	0
3	10	20	15	5	5	5
4	20	5	10	5	5	10

Example: results plot



Example: replacing content, footnotes, external figure

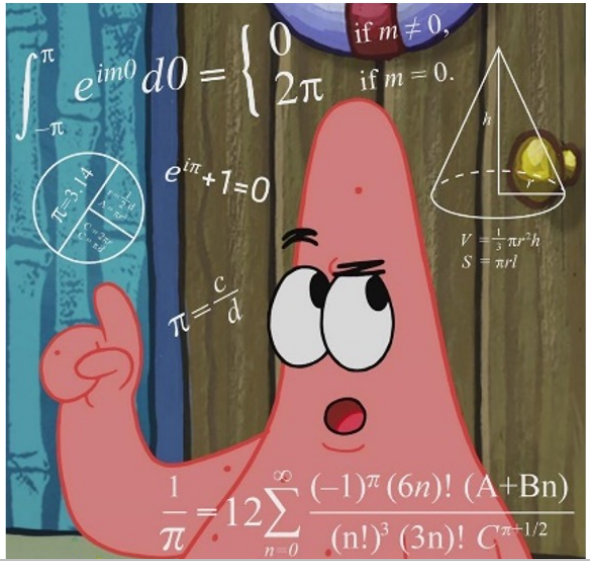
- first point of the first group
- second point, with a key word
- third point



Example: replacing content, footnotes, external figure

- a figure from a source: $13 trillion[a]
- another figure: 29.3 bln[b]
- a third figure: $2.7–6.2 trillion[c]

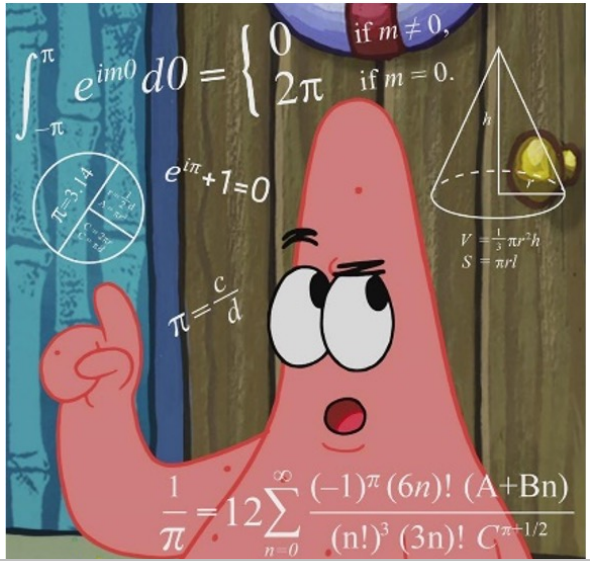
[a] Source: Source A
[b] Source: Source B
[c] Source: Source A



Example: replacing content, footnotes, external figure

- a point with a key word
  - a sub-point

$\Longrightarrow$ conclusion in one line



Example: title of step 1

- highlighted at steps 1 and 3, range kept in a macro
- highlighted only at step 2
- the same two ranges, written directly

a TikZ node highlighted at steps 1 and 3

Example: the title changes with the step

- highlighted at steps 1 and 3, range kept in a macro
- highlighted only at step 2
- the same two ranges, written directly

a TikZ node highlighted at steps 1 and 3

Example: the title changes with the step

- highlighted at steps 1 and 3, range kept in a macro
- highlighted only at step 2
- the same two ranges, written directly

a TikZ node highlighted at steps 1 and 3

Example: one figure, first frame



Example: one figure, first frame



Example: one figure, first frame



Example: the same figure, second frame



Example: the same figure, second frame



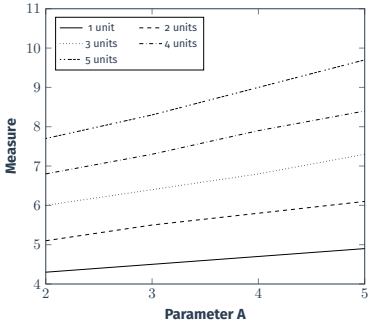
Example: completed sentence, line plots, summary line

Measure varying the number of

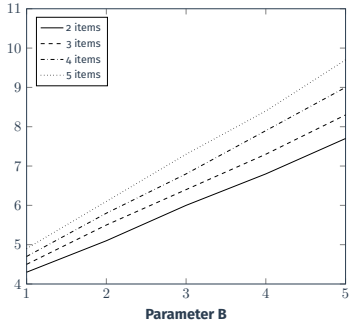
- parameter A in $\{2, ..., 5\}$
- parameter B in $\{1, ..., 5\}$

Example: completed sentence, line plots, summary line

Measure varying the number of parameter A and parameter B



Varying parameter A



Varying parameter B

$\Longrightarrow$ linear increase of the measure

Example: stacked tables with captions

Name	Column 2	Column 3	Description
Item1	A	B–C	First item
Item2	A	B	Second item
Item3	C	C	Third item

First table

Name	Column 2	Column 3	Description
Item4, Item5	A	B	A longer description of both items
Item6	C	B	Another long description that wraps on two lines

Second table

Example: table next to bullet points

Step	Value	Cost	
		Unit A	Unit B
Step 1: first	8,932,049	0.41	1,288.920
Step 2: second	5,136,015	0.235	741.141
Step 3: third	4,076,328	0.187	588.225
Total	18,144,392	0.832	2,618.286

- Cost=*value*×*unit cost*
- Unit cost: 45.85×10^{-9}
- 1 unit A = 3,147.28 unit B
- a note in *italics*

Example: title slide with logos and funding

Speaker[1] Author Two[1] Author Three[2]

[1]Affiliation 1

[2]Affiliation 2





The work was partially supported by *i)* project A, funded by the funder under the call number 1; *ii)* project B, funded by another funder.

Example: chevrons, fading background

A sentence introducing the chevrons...



Example: chevrons, fading background

A sentence introducing the chevrons...



Example: chevrons, fading background

A sentence introducing the chevrons...



- **Item 1**: first short statement
- **Item 2**: second short statement
- **Item 3**: third short statement

Example: chevrons, fading background

A sentence introducing the chevrons...



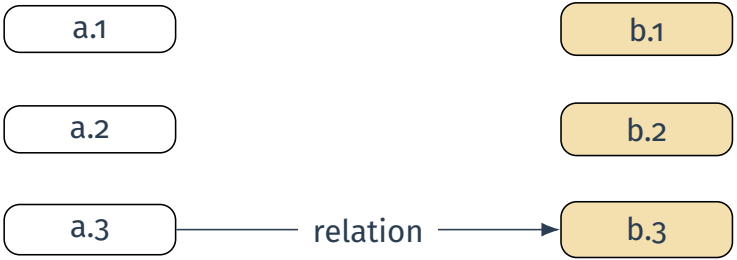
Example: reusable component with `\pic`



Example: reusable component with `\pic`



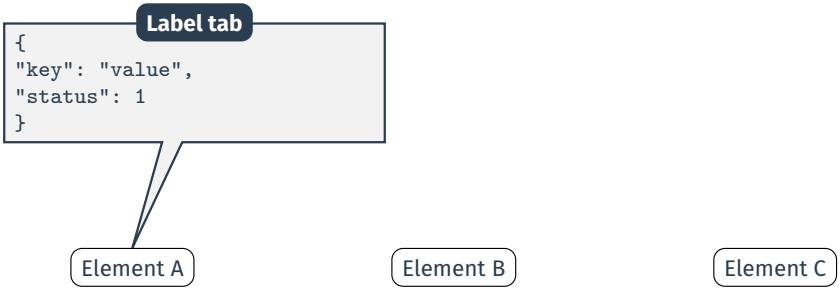
Example: reusable component with `\pic`



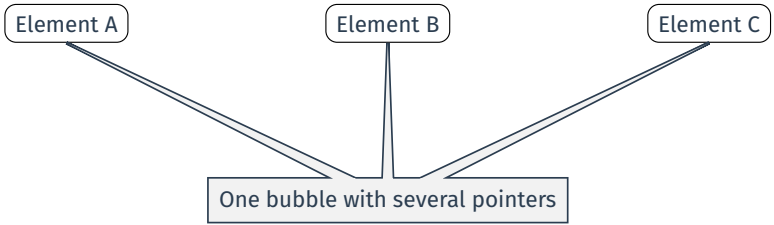
Example: reusable component with `\pic`



Example: callouts, code, several pointers, characters



Example: callouts, code, several pointers, characters



Example: callouts, code, several pointers, characters

Element A

Element B

Element C

